

$\hat{x} \in [0, 1]$

logical grid

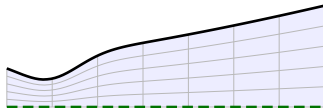
fixed: topology, connectivity, boundary tags

$$x = \hat{x} L$$

$$y = r y_{\text{wall}}(x; \vec{a})$$



the *only* place the
design variables $\vec{a}$ enter



x [m]

physical mesh

moves with the design

Splitting the mesh this way is what makes shape derivatives practical. The integer connectivity is built once and treated as constant; only the node coordinates — and therefore the Jacobians, normals and mass matrices derived from them — depend on $\vec{a}$. Those are written through an injected array module, so `jax.grad` propagates straight through the mesh motion into the residual.